

ISY122 – Analytics Techniques and Tools

PREDICTING STUDENT EMPLOYABILITY USING MACHINE LEARNING BASED ON SOFT SKILLS ASSESSMENT

ABDULKARIM, HAMAD D.
MACARANDAS, ROHAIMAH S.

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INTRODUCTION

- Philippine unemployment reached 5.8% in January 2026, affecting nearly 3 million Filipinos
- This study uses machine learning to predict student employability based on soft skills

RESEARCH QUESTIONS

1. Is there a significant relationship between soft skills and student employability?
2. Which soft skills significantly influence the prediction of student employability?
3. Which data mining algorithm provides the highest accuracy in predicting student employability?

DATA

Source

KAGGLE
open source platform

Key Variables

EIGHT (8) INDEPENDENT VARIABLE

General Appearance
Manner of Speaking
Physical Condition
Mental Alertness
Self Confidence
Ability to Present Ideas
Communication Skills
Student Performance Rating

ONE (1) DEPENDENT VARIABLE

CLASS: Employable, LessEmployable

MEASURED USING A 5-POINT LIKERT
SCALE

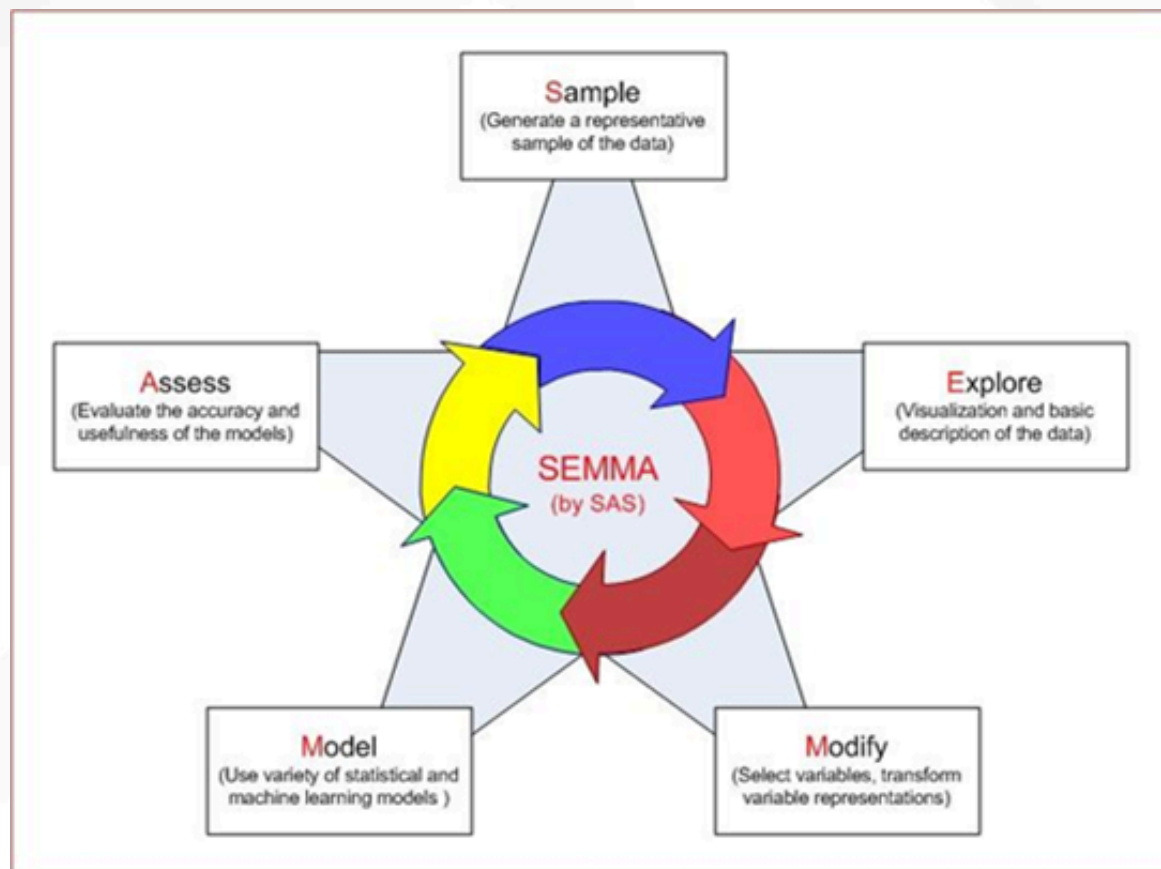
Data Size

2,982
student records

METHODOLOGY



SEMMA FRAMEWORK



MACHINE LEARNING MODEL

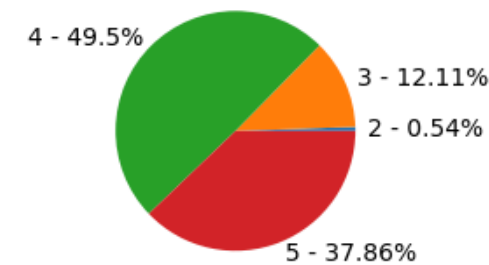
● DT ● SVM ● KNN



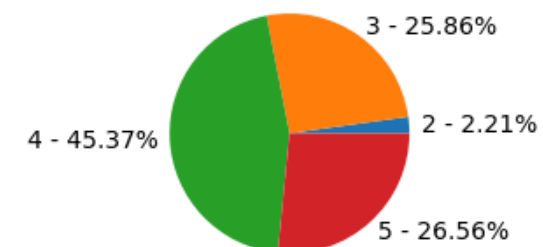
Exploratory Data Analysis

Population Pie Chart by Features

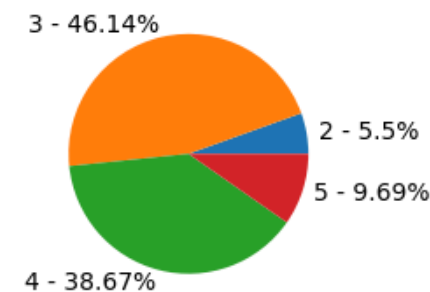
Population General Appearance



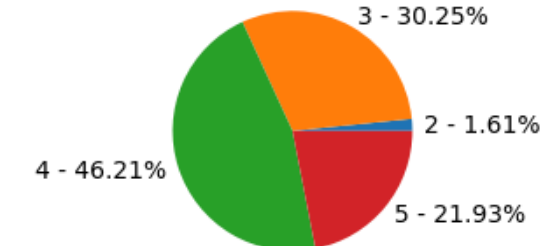
Population Mental Alertness



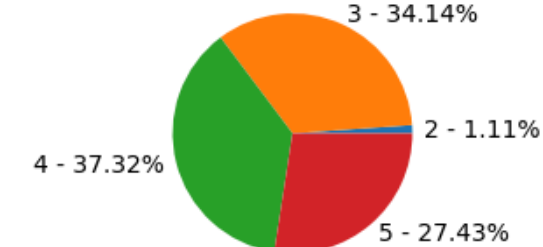
Population Communication Skills



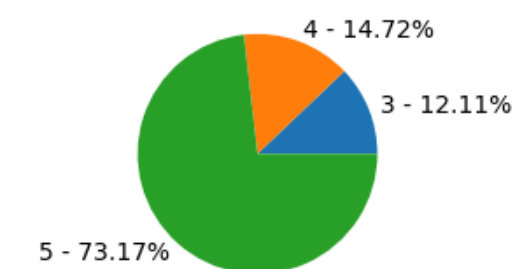
Population Manner Of Speaking



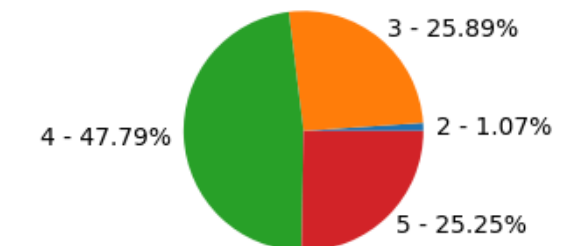
Population Self-Confidence



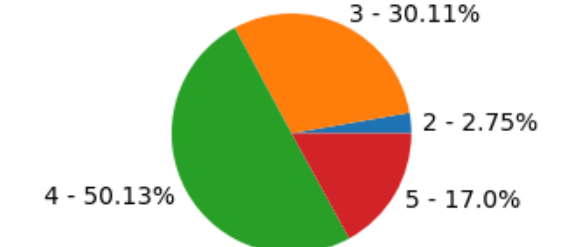
Population Student Performance Rating



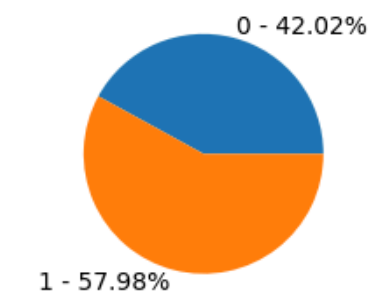
Population Physical Condition



Population Ability To Present Ideas

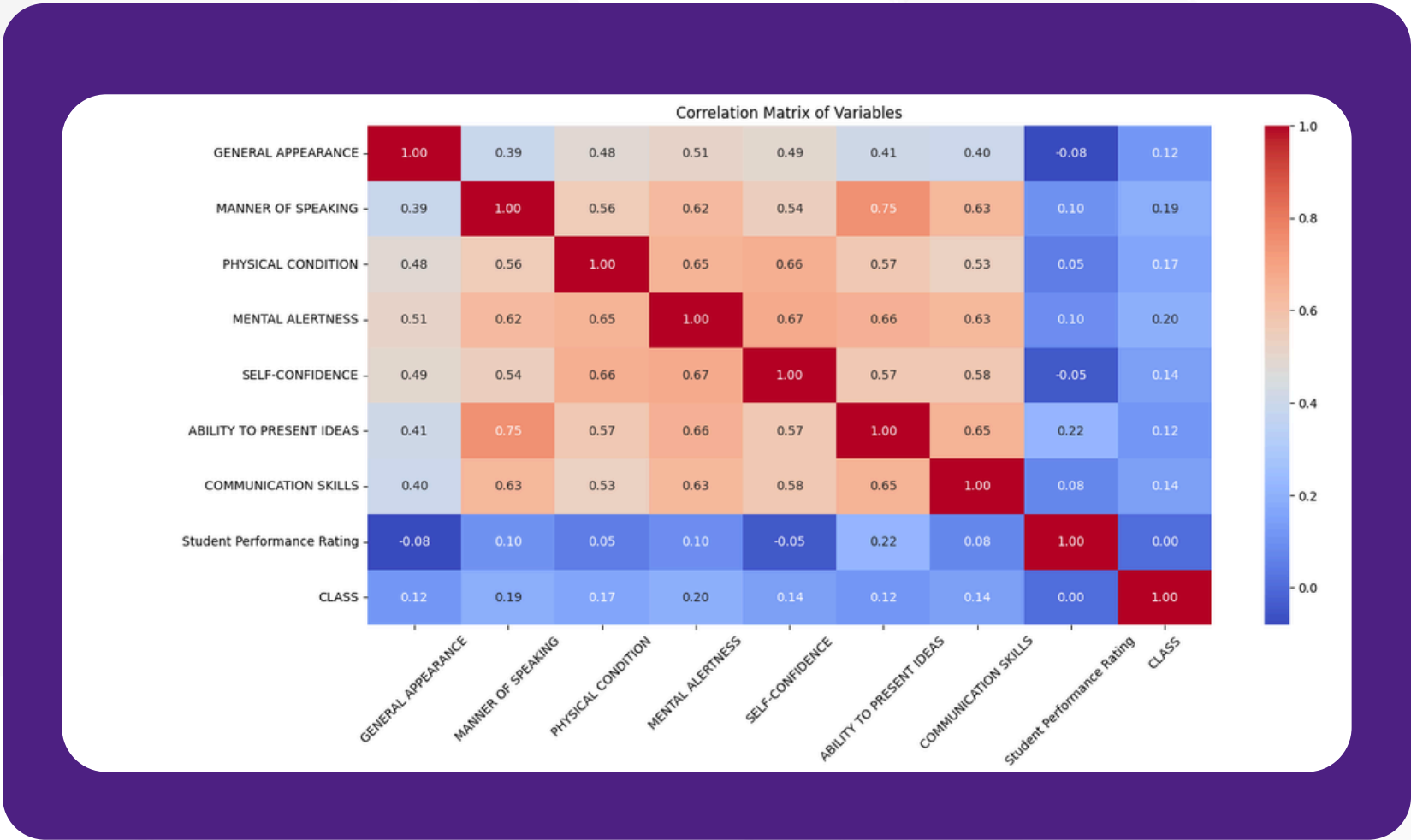


Population Class

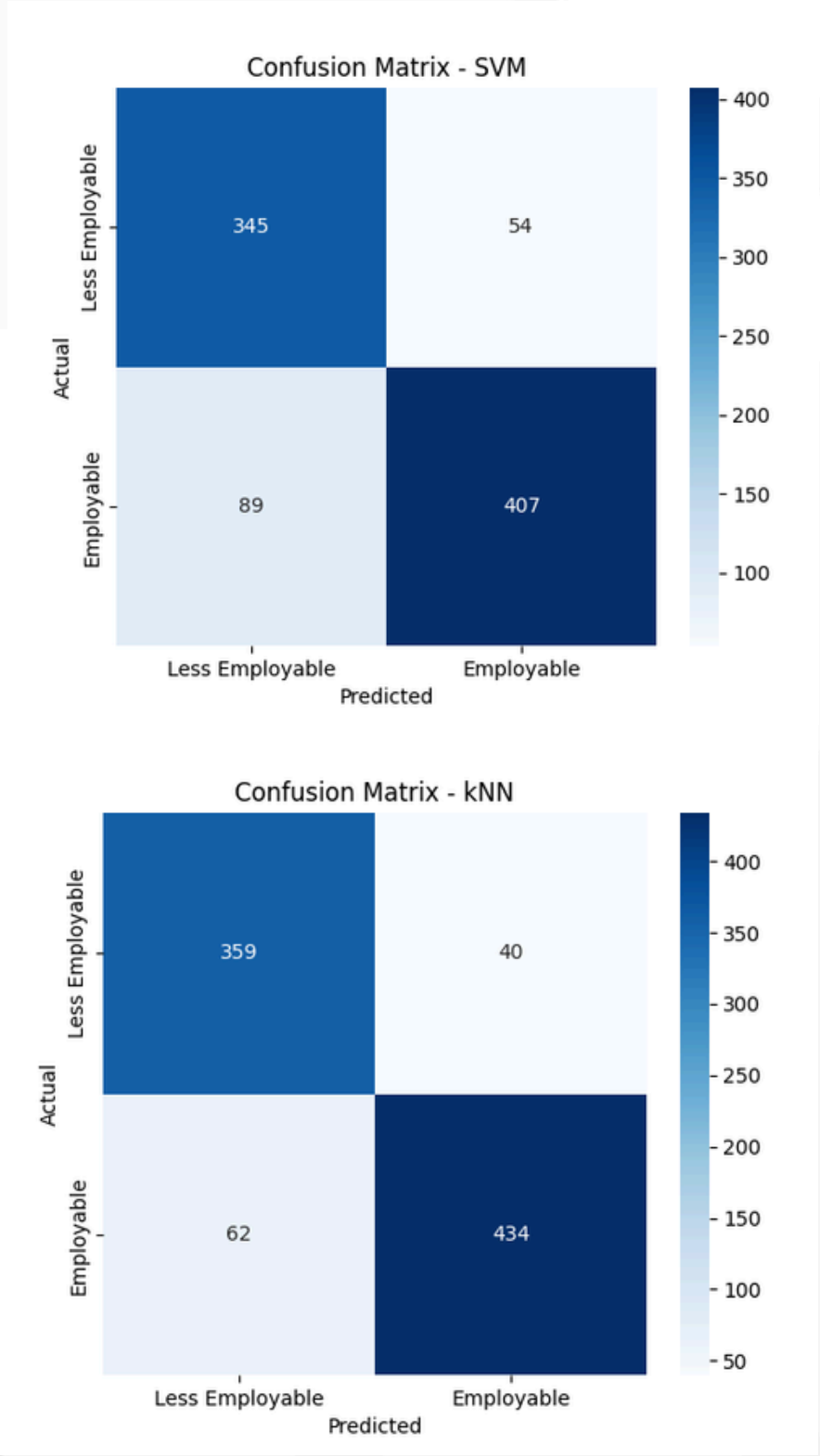
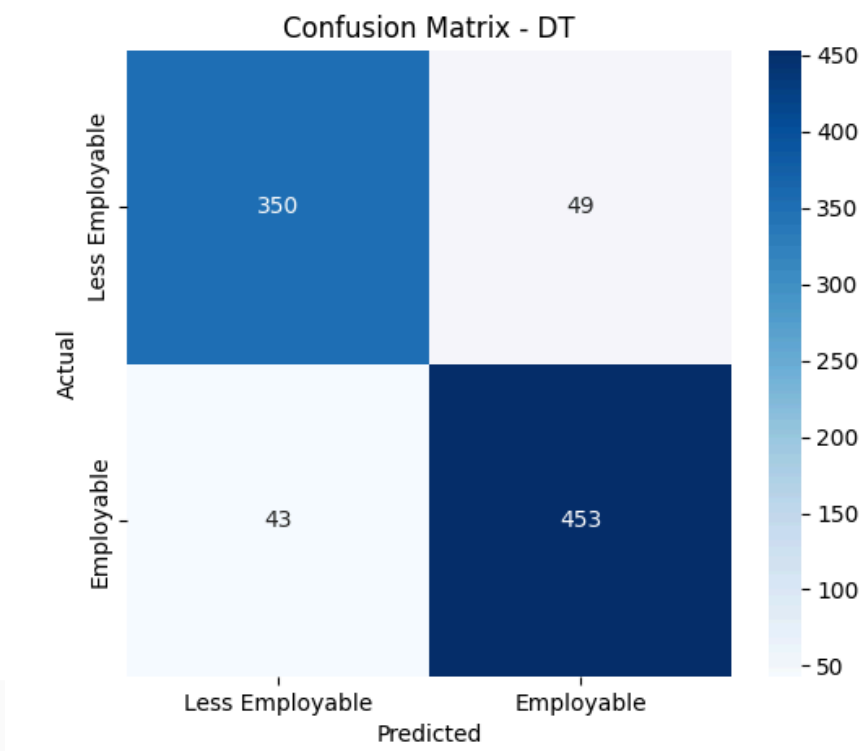


KEY RESULTS

Correlation Matrix

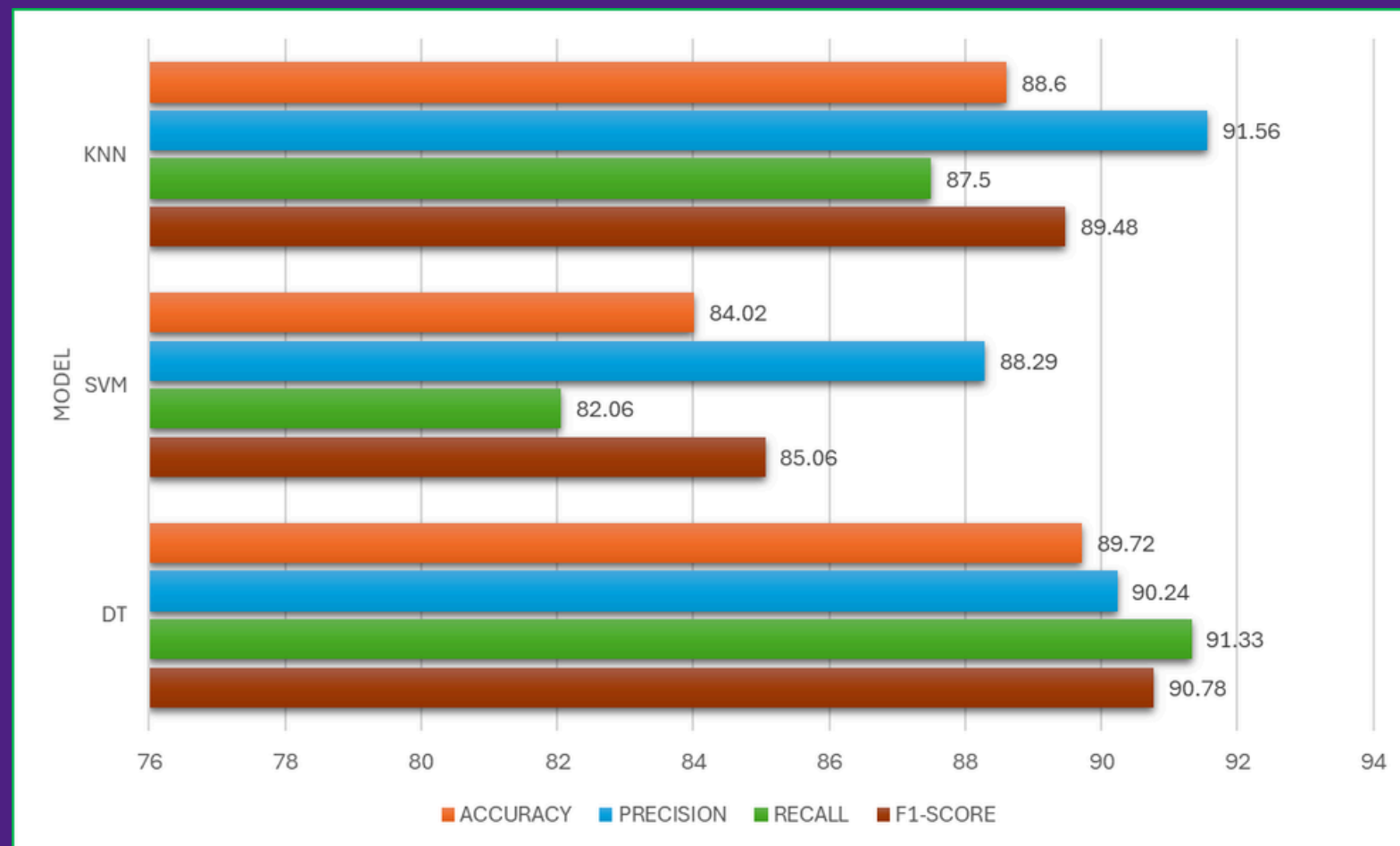


Confusion Matrices

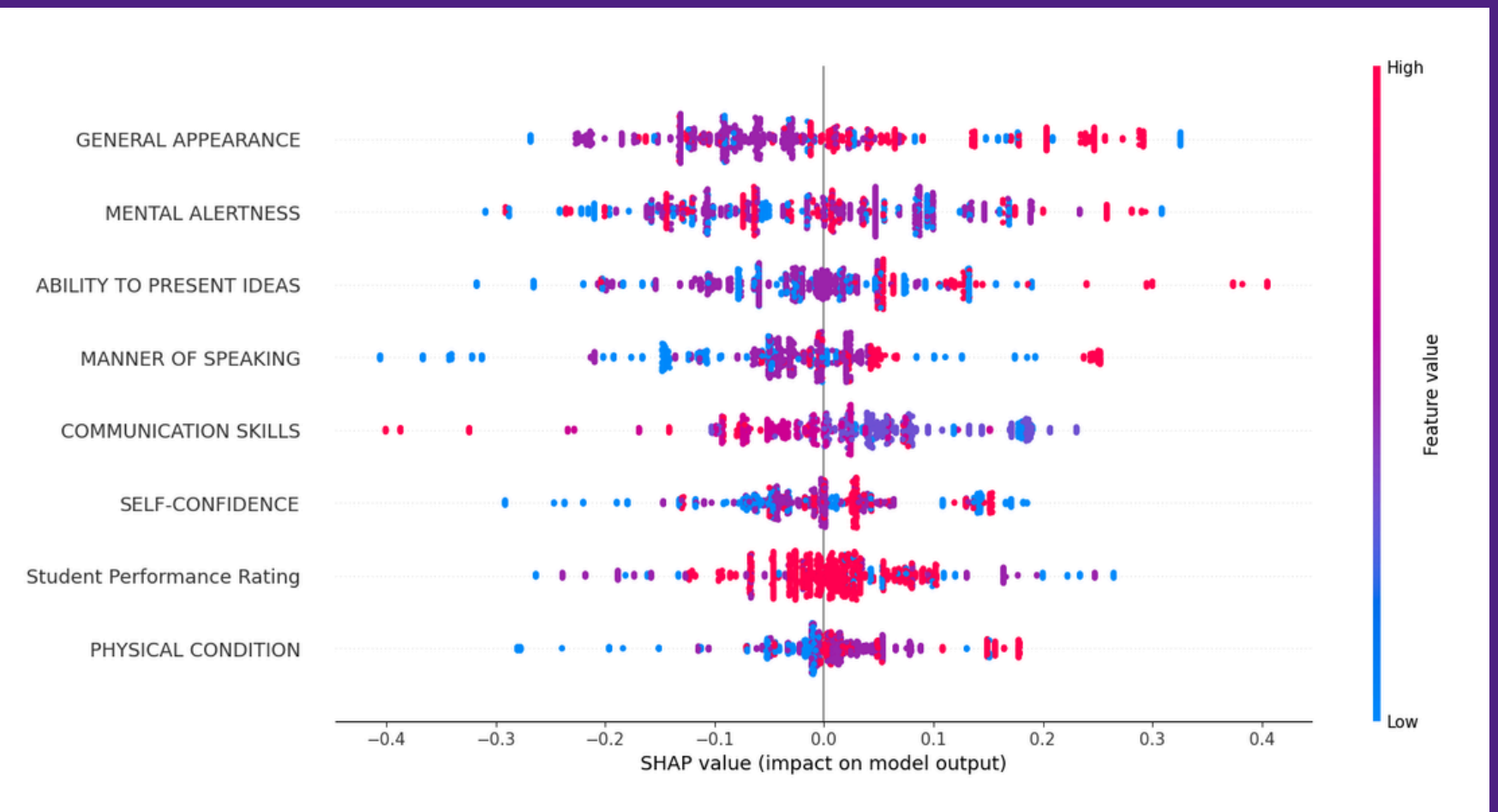


KEY RESULTS

Assessment Scores for Each Model



SHAP Model



INSIGHTS & DISCUSSION

- There is a significant relationship between soft skills and student employability.
- General Appearance, Mental Alertness, and Ability to Present Ideas were the strongest predictors of employability.
- Communication Skills and Manner of Speaking also highly influenced employability prediction.
- Academic performance and physical condition had lower impact compared to soft skills.
- Findings suggest that employers value communication, presentation, and cognitive skills more than academic-related factors.
- Decision Tree achieved the highest accuracy (89.72%), followed by KNN (88.60%) and SVM (84.02%). Decision Tree showed the most balanced and reliable classification results.
- Machine learning techniques can help educational institutions identify skill gaps and improve student preparedness for employment.

LIMITATIONS

The study is limited by the use of secondary self-assessment data, focus on soft skills only, and the use of only three machine learning models. Other factors and advanced algorithms may improve prediction performance.

FUTURE WORKS

Future studies may include additional factors such as technical skills, internships, certifications, and academic performance, while also testing more advanced machine learning models. Researchers may also use real-time data and develop systems that help schools assess and improve student employability.



THANKYOU!